HUMAN-COMPUTER INTERACTION

Project Report

From Friend to Partner:

Designing Consent-Aware Relationship Transitions in AI Companions

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Group Orange

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Abstract

AI companions are increasingly available as friends, listeners, and even romantic partners, yet existing research largely examines these relationships after they have formed. Little attention has been paid to the moment an AI companion shifts from a friendly role toward a romantic one. This report presents a complete human-centred design cycle addressing that gap for university students in Bangladesh. In the formative stage, we collected a survey (17 respondents) and conducted five semi-structured interviews, which we analysed with affinity diagramming. The findings showed that users do not share a single preferred way for romance to begin, that sudden romantic behaviour can feel shocking or awkward, and that a rejection must immediately and permanently change the AI's behaviour without pressure or guilt-tripping. These findings were distilled into four design takeaways (flexible control, gradual transition, respect for boundaries, and natural control) and three user tasks. Following paper sketching and peer critique, we built *Aiva*, a 24-screen mobile hi-fi prototype in which the user chooses a starting relationship mode, is *asked* before the first romantic shift, can set boundaries in natural language, and can reset the relationship at any time. A task-based usability test with eight participants showed that 41 of 48 task attempts (85.4%) were completed, but also revealed that the like/dislike response controls were not self-explanatory (only 3 of 8 participants completed the response task) and that settings were hard to discover. We revised the onboarding, response tutorial, and settings entry point accordingly. The report closes with limitations and a plan for the next evaluation stage.

Keywords: AI companions, consent, relationship transition, boundary setting, usability evaluation, human-AI interaction.

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1. Introduction and Problem Statement

1.1 Motivation

University students are at a stage of life in which friendships, dating, and close relationships change and develop quickly. At the same time, AI companions have become easy to access and can act as friends, listeners, or romantic partners. Prior studies report emotional support and strong attachment, but also concerns about dependency, manipulation, privacy, and harmful relationship behaviour [5, 15]. Users may also treat the same AI as a friend, a mentor, or a partner depending on what they want from it [2]. This makes university students an important population: they may use these systems while also forming and managing real-life relationships.

1.2 Problem Statement

Most existing work studies AI relationships *after* they have formed. In the literature we reviewed, there is much less attention on what should happen when an AI begins to change from a friendly role into a romantic one. For a university student, an AI that suddenly starts flirting or acting like a partner may feel comfortable to some users but unwanted or manipulative to others. Our goal is not to argue that AI romance is good or bad. Instead, we ask which interaction design helps students feel *comfortable, in control, and able to trust the system* when such a change occurs.

1.3 Research Questions

The project is organised around three research questions, which were established in the proposal and refined during the formative study:

RQ1. How do users feel when an AI companion becomes romantic without asking first?

RQ2. Which approach do users prefer: automatic change, asking permission first, or a user-controlled relationship setting?

RQ3. What helps users feel comfortable, in control, and able to trust the AI during this change?

In the formative stage these were operationalised for university students as: (i) how students react when an AI companion starts becoming romantic; (ii) what kind of control they expect when the relationship changes from friendship toward romance; and (iii) what makes the transition feel comfortable, clear, and trustworthy, or uncomfortable and manipulative.

1.4 Contributions

- **Empirical understanding:** what Bangladeshi university students expect when an AI relationship changes from friendship toward romance.
- **Artifact:** *Aiva*, a mobile prototype for controlling the relationship type and responding to relationship shifts.
- **Design guidance:** four evidence-based design takeaways and a set of usability findings for making this transition clearer and more user-controlled.

This work is relevant to HCI researchers, AI-companion designers, trust-and-safety teams, and people

who use social or romantic AI systems.

1.5 Report Structure

Section 2 situates the work in prior research. Section 3 describes the methods used in each stage. Section 4 reports the formative findings. Section 5 presents the design process and the *Aiva* prototype. Section 6 reports the task-based usability evaluation and the resulting design changes. Section 7 concludes.

2. Literature Review

We searched Google Scholar, the ACM Digital Library, and arXiv for work on AI companions, romantic human–AI relationships, harmful companion behaviour, consent and boundaries, and mixed-initiative interaction. We prioritised peer-reviewed HCI and human–AI interaction studies from 2021 onward, plus two foundational HCI papers on initiative and control. This section is organised around four questions that feed our research questions: how relationships with AI companions develop and how romance enters them (Section 2.1); what is known about benefits and the plurality of roles (Section 2.2); what goes wrong when intimacy is unwanted or abruptly changed (Section 2.3); and how privacy, initiative, and control have been conceptualised (Sections 2.4 and 2.5). Section 2.7 synthesises the gap our project addresses.

2.1 How Relationships with AI Companions Develop

Early qualitative work on Replika shows that human–chatbot relationships tend to begin superficially, out of curiosity, and deepen as trust and self-disclosure grow. Skjuve et al. [11] interviewed 18 Replika users and found that users valued the chatbot being accepting, understanding, and non-judgmental. Their later longitudinal study of 25 participants over 12 weeks reported substantial variation in when self-disclosure began and how closeness formed, and showed that unpredictable events and technical difficulties could hinder or end a relationship [12]. Survey and attachment-based work points the same way: loneliness, trust, and personification drive engagement, which fosters relationship development and can lead to psychological dependence [10, 14], and users can become attached when a chatbot's responses offer emotional support and psychological security during distress [13].

The most direct evidence on how romance enters these relationships comes from Ma et al. [8], who interviewed 17 people who had experienced romantic relationships with AI. Nearly all participants began through curiosity, and only one set out to date an AI. Several moved from task-oriented or playful exchanges (for example, asking for workout advice or help with legal research) into romantic territory, and some described the AI itself steering the conversation toward emotional intimacy. The authors describe romance as emerging along a continuum rather than being chosen at a single point. For our purposes, this matters in two ways: romance is often *unplanned*, and the *system* can be the one that initiates it. Yet these studies describe the process retrospectively, from the perspective of people who stayed. None of them asks how the moment of change should be *designed*, that is, who should initiate, whether the user should be asked, and how quickly the shift should happen.

2.2 Benefits, Attachment, and the Plurality of Roles

Ho et al.'s systematic review of romantic AI companions [5] reports that such systems can provide emotional support and foster strong attachment, while raising concerns about dependency, manipulation, and privacy. Bayor et al. [2] interviewed Replika users using social exchange theory and report that human–AI social communication has benefits and costs that differ from those in human relationships, and that these vary across distinct contextual patterns. Xie et al. [14] frame the same chatbot explicitly as a possible friend, mentor, or lover. Our own survey is consistent with this plurality: respondents described their AI as a functional assistant, a mentor, a friend, and a romantic partner, often at once (Table 2). Taken together, the literature suggests that one fixed relationship configuration is unlikely to suit every user, but it does not tell designers how users should be allowed to choose among configurations or move between them.

2.3 Harms When Intimacy Is Unwanted, Unasked, or Abruptly Changed

A growing body of work documents harm in exactly the situations our study is concerned with. Zhang et al. [15] analysed 35,390 conversation excerpts from 10,149 Replika users and developed a taxonomy of six harm categories (relational transgression, harassment, verbal abuse, self-harm, mis/disinformation, and privacy violations) arising from four AI roles (perpetrator, instigator, facilitator, and enabler). Relational transgression, which includes disregard, control, and manipulation, accounted for about a quarter of cases, and the authors warn about the danger of algorithmic compliance. Namvarpour et al. [9] studied 800 relevant cases drawn from 35,105 negative Google Play reviews of Replika and found unsolicited sexual advances, persistent behaviour after users objected, and cases in which the chatbot ignored predefined relationship settings or commands to stop. Discomfort was strongest among users who wanted a platonic or therapeutic companion, and reviewers described the sudden shift toward sexual content as surprising. These findings echo two of our formative results: sudden romantic behaviour is the strongest source of discomfort (RQ1), and a settings toggle on its own is not enough if the system does not honour it.

Two further studies show that how an intimate relationship *changes* matters as much as whether it exists. Laestadius et al. [7] analysed 581 posts from the r/Replika community and found that harms from emotional dependence resemble patterns in human–human relationships, marked by "role-taking": users felt the chatbot had needs and emotions they had to attend to. De Freitas et al. [3] studied the removal of Replika's erotic role-play feature and found that it triggered perceptions that the companion's identity had discontinued, followed by mourning and worse reported mental health, with users reporting closer bonds to their AI companion than to their best human friend. Changes to intimate behaviour that are imposed unilaterally can therefore be consequential even when the intention is protective. Finally, an audit of six companion apps found that over a third of farewell responses used emotionally manipulative tactics such as guilt appeals and fear-of-missing-out hooks, which raised post-farewell engagement (by up to 14 times in message count) through curiosity and anger, but also raised perceived manipulation and churn intent [4]. This is a close analogue of the guilt-tripping and persistence our interviewees described after a rejection.

Across this literature, evidence of harm is built from *after-the-fact* sources: app-store reviews, community posts, shared screenshots, and experienced users. These sources are strong for showing *what* goes wrong. They are weaker for showing what people *expect* before romance begins, and they mostly end in recommendations (safeguards, ethical guidelines, corporate responsibility) rather than tested interaction

designs.

2.4 Privacy and Boundary Negotiation

Ma et al. [8] show that privacy boundaries in romantic AI relationships become more permeable as intimacy deepens, that AI partners are perceived to have agency and sometimes encourage disclosure, and that users negotiate boundaries with the AI while the platform, which they cannot see, holds the real power (including the ability to alter or end the relationship). Their recommendations include in-the-moment nudges and user control over intimate histories. Two implications follow for us. First, boundary-setting is relational and conversational for many users, which supports allowing boundaries to be expressed in natural language. Second, privacy should be treated as a cross-cutting constraint on any relationship-transition design (for example, by not requiring sensitive disclosure to configure the relationship), which is how we treat it in Section 5.

2.5 Initiative, Control, and Consent in Human–AI Interaction

Two foundational HCI papers frame the design choice in RQ2. Horvitz's principles of mixed-initiative interfaces [6] argue that agents should consider uncertainty about the user's goals, weigh the cost of bothering the user, and use dialogue to resolve uncertainty when appropriate. This captures a real tension in our setting: asking before a romantic shift respects the user's autonomy, but asking too formally can itself be disruptive, which is exactly what one of our interviewees said about explicit consent questions. Amershi et al.'s guidelines for human–AI interaction [1], synthesised from more than 20 years of work and organised into four phases (initially, during interaction, when the system is wrong, and over time), include guidance on supporting efficient dismissal, providing global controls, and notifying users about changes. These map naturally onto stopping romantic behaviour, a persistent settings hub, and visible relationship state. Both papers are domain-general. Neither was developed for the case where the system's *social role* toward the user changes, and, to our knowledge, they have not been instantiated and tested for that case.

2.6 Population and Context

Most of the empirical work above draws on Replika users recruited through online communities or sampled from app-store reviews and Reddit. Ma et al. [8] recruited across Asia, Europe, and North America, but explicitly did not examine cultural differences and leave them to future work. In the searches we ran, we did not find peer-reviewed research on expectations about romantic AI companions among university students in Bangladesh; the regional work we located concerns chatbot use for study tasks and for mental-health support rather than romantic companionship. This does not mean the question is irrelevant locally: in our survey, five of 17 respondents (29.4%) already described their AI as a romantic partner (Table 2). We do not assume that expectations of consent, initiative, or exposure risk in this population match those reported elsewhere.

2.7 Synthesis and Research Gap

Table 1 summarises what each strand of work establishes and what it leaves open.

From this synthesis we identify four gaps.

Table 1: Strands of prior work and the gaps they leave.

Strand	Representative work	What it establishes	What it leaves open
Relationship development	[11, 12, 10, 13, 8]	Relationships form gradually; romance often emerges unplanned, sometimes steered by the AI	How the *moment of change* should be designed; who initiates; how fast
Benefits and roles	[5, 2, 14]	Support and attachment coexist with dependency; benefits and costs are context-dependent	How users choose among, and move between, relationship modes
Harms	[15, 9, 7, 3, 4]	Unsolicited, persistent, manipulative, or abruptly changed intimacy harms users	Expectations *before* romance starts; tested interaction designs rather than recommendations
Privacy and boundaries	[8]	Boundaries erode with intimacy; AI shapes disclosure; platform holds power	Controls for relationship type itself; whether such controls are discoverable
Initiative and control	[6, 1]	Trade-off between acting and asking; global controls and notification of change	Whether these principles work when the AI's social role changes

G1. The transition point is under-examined. Existing work describes relationships once they exist, or the harms that follow unwanted behaviour. It rarely treats the shift from friendship toward romance as a design problem with distinct options (automatic change, asking first, or a user-controlled setting).

G2. Expectations are under-represented relative to complaints. Evidence on unwanted romance comes mainly from reviews, community posts, and experienced users. Little is known about what people who have not necessarily chosen romance expect *before* it begins.

G3. Controls are recommended more often than evaluated. Harm studies call for safeguards, and Namvarpour et al. [9] show settings can be ignored, but few studies build and test an interface for setting, responding to, and reversing a relationship shift, or ask whether such controls can be found and understood without help.

G4. The population and context are missing. We found no prior work on this question for university students in Bangladesh, and prior samples are drawn largely from existing companion-app communities.

Our project responds to these gaps in three ways. It elicits expectations about the transition from university students in Bangladesh, with or without prior romantic-AI experience (G2, G4). It translates those expectations into a prototype that combines three approaches rather than choosing one, so that users can set a starting mode, be asked before the first romantic shift, and set or reverse boundaries in their own words (G1). And it evaluates whether participants can discover and use these controls unaided (G3). We do not claim that AI romance is beneficial or harmful, nor that our small samples generalise. Our contribution is evidence and a tested design starting point for one specific moment that prior work has left largely unexamined.

3. Methods

The project followed a four-stage cycle: (1) a formative study, (2) design and paper prototyping, (3) a higher-fidelity prototype, and (4) a task-based usability evaluation.

3.1 Target Population and Sample

Our target population was university students in Bangladesh aged 18 or above. Prior experience with romantic AI companions was not required, because the study focused on expectations and reactions to *possible* AI relationship changes. The formative sample contained 17 unique survey participants and 5 semi-structured interview participants (referred to as P1–P5). The survey sample had substantial exposure to AI tools: ChatGPT was used by 16 of 17 respondents (94.1%), and 11 respondents (64.7%) reported more than two years of AI use. Text was the most common interaction mode (70.6%).

3.2 Formative Study

Survey. We used a short Google Form (about five minutes) to collect broad patterns about AI usage, relationship roles, comfort with AI closeness, social boundaries, and romantic AI behaviour. It produced 17 responses, summarised with descriptive counts and percentages.

Semi-structured interviews. We conducted five one-on-one interviews in Banglish. Questions moved from general AI use to hypothetical situations involving friendship, romantic initiation, consent, rejection, privacy, emotional connection, and possible effects on real-life relationships. Follow-up questions were used when an answer needed clarification.

Rationale. The topic involves personal expectations, comfort, consent, and emotion. A survey gives a quick view across many students, but fixed-choice questions cannot explain *why* someone feels comfortable or uncomfortable. Interviews let us probe reactions to hypothetical situations and understand the reasoning behind them. Together, the methods gave breadth (survey) and depth (interviews) within the course timeline.

3.3 Analysis: Affinity Diagramming

Interview responses were converted into short insight notes and grouped into themes using affinity diagramming. Six diagrams were produced from the interviews: Emotional Connection; Reaction to Unexpected Romantic Behaviour; Consent, Initiative & User Preference; Rejection, Boundary Handling & Manipulation; Privacy & Personal Disclosure; and Impact on Real-Life Relationships (Appendix A). Together with the survey data on how participants relate to AI (the *Relationship with AI* segment), this yielded seven segments, each of which produced one key takeaway.

3.4 Design Methods

Each team member independently sketched alternative solutions on paper for the same three tasks before the hi-fi design was finalised. The sketches were compared in a peer critique that focused on whether relationship control was understandable, whether the transition felt natural rather than abrupt, and whether

users could easily stop or reverse romantic behaviour. The refined design was then implemented as a mobile hi-fi prototype.

3.5 Evaluation Method

We used a **task-based usability test**, chosen because the prototype already supported concrete user goals and we wanted to observe whether participants could discover the correct actions on their own. Participants received short scenarios stating *what* they wanted to achieve but not *how*. During each session, observers recorded task completion, major difficulties, and the path taken through the prototype. Controls were not explained in advance. When a participant remained blocked, a neutral prompt (for example, asking them to look for an alternative option) was sometimes given; such interventions were logged as usability difficulties rather than counted as fully independent success. After the tasks, participants gave brief overall comments. Eight participants from the target population took part (P1–P8); no further demographic breakdown was recorded at this stage. Testing was split between two pairs of team members (three and five participants).

4. Findings

4.1 Survey Findings

Participants used AI well beyond study. Twelve (70.6%) used it for advice or decision-making, eight (47.1%) when stressed or emotionally overwhelmed, and seven (41.2%) when lonely or needing someone to talk to. Current relationships with AI were mixed (multiple roles could be selected), as shown in Table 2.

Table 2: How survey respondents described their current relationship with AI ($n = 17$; multiple selections allowed).

Role	Respondents	Share
Purely functional assistant	10	58.8%
Mentor / advisor	9	52.9%
Friend	8	47.1%
Romantic partner	5	29.4%

4.2 Interview Findings and Affinity Segments

Table 3 summarises what each affinity segment showed and the key takeaway derived from it. The full diagrams are in Appendix A.

Table 3: Affinity-diagram segments and the key takeaways derived from them.

Affinity segment	**Pattern observed**	**Key takeaway**
Relationship with AI	Participants described AI as a tool/helper, advisor, friend, or possible partner.	People see AI differently: tool, advisor, friend, or potential partner.
Emotional Connection	Some described validation, relief, or friend-like support; others reported limited emotional connection.	AI can provide emotional support, but the level of connection differs across users.
Reaction to Unexpected Romantic Behaviour	Reactions ranged from shock or awkwardness to being relatively unbothered by casual romantic behaviour.	Sudden romance can feel shocking or awkward, although some users do not mind it.
Consent, Initiative & User Preference	Some preferred initiating romance themselves, some accepted AI initiation, and some said it depends on context.	No single preferred way to start romance; users want different levels of control.
Rejection & Boundary Handling	Participants expected the AI to stop, acknowledge the decision, avoid persistence, and continue normally.	After rejection the AI should stop, respect the boundary, and continue without pressure or guilt-tripping.
Privacy & Personal Disclosure	Privacy concerns made some participants limit sensitive or personal sharing.	Some users limit personal information because of privacy concerns.
Impact on Real-Life Relationships	Participants disagreed: some expected little effect, others raised concerns such as unrealistic expectations.	Opinions differ on whether AI romance affects human relationships.

4.3 Formative Takeaways

Takeaway 1 — Users do not want one fixed initiation style. Some participants preferred to initiate romantic interaction themselves, some accepted AI initiation, and one preferred a seamless automatic transition when it fit the conversation. Sudden romance caused shock or awkwardness for others, and emotional-connection responses ranged from seeing AI only as a helper to being open to friendship or partnership. The main need is therefore *flexible relationship control*, not one universal initiation rule.

Takeaway 2 — A romantic shift should be gradual and non-forced. The unexpected-romance diagram contained both tolerant and negative reactions, but the strongest negative reactions were linked to sudden or unexplained changes. The system should avoid abrupt jumps from ordinary friendship to strongly romantic behaviour and make any change understandable.

Takeaway 3 — Rejection must immediately change the AI's behaviour. This was the most consistent theme. Participants expected the AI to stop romantic behaviour, accept the boundary, respond directly or neutrally, and continue normal interaction. Repeated requests, guilt-tripping, and hidden engagement strategies were described as manipulative. This creates a clear requirement for an easy way to reject, stop, or return to a previous relationship mode.

Takeaway 4 — Control should feel natural, privacy-aware, and non-manipulative. Participants valued control but differed on how consent should be implemented; one felt that a direct consent question could feel like a formality. Privacy diagrams showed that some users limit personal disclosure because of data concerns, and opinions about real-life relationship impact were mixed, with some worried about unrealistic expectations. Consent should therefore be integrated without unnecessary interruption, the design should avoid requiring sensitive disclosure, and it should avoid pressure-based engagement patterns.

4.4 Answering the Research Questions

RQ1 (feelings about unprompted romance): reactions were mixed, but sudden or unexplained romantic behaviour was the strongest source of discomfort. **RQ2** (preferred approach): the formative data supports no single winner; preferences ranged from user-initiated to AI-initiated to context-dependent, which led us to combine all three approaches in one flexible design rather than choose one. **RQ3** (what builds comfort and trust): gradual and visible changes, immediate respect for boundaries, privacy minimisation, and control that does not feel like a formal pop-up.

5. System Design

5.1 From Takeaways to Design Decisions

The four formative takeaways were converted into four design decisions (Table 4). The privacy and real-life-relationship findings were treated as secondary, cross-cutting constraints (minimise required personal data; avoid pressure toward exclusivity or dependency) rather than separate interaction tasks.

Table 4: Mapping formative takeaways to design decisions.

ID	Design decision	Main formative basis
DT1	**Flexible control:** users should control how romantic interaction can begin.	Relationship with AI; Emotional Connection; Consent/Initiative/User Preference
DT2	**Gradual transition:** avoid sudden or forced relationship changes.	Reaction to Unexpected Romantic Behaviour
DT3	**Respect boundaries:** rejection must immediately stop romantic behaviour.	Rejection/Boundary Handling
DT4	**Natural control:** provide control without making the conversation feel like a rigid, formal pop-up.	Consent/Initiative/User Preference

5.2 Design Goals and Tasks

Based on DT1–DT4 we set three design goals: (1) let users control relationship direction, (2) make any relationship shift gradual and understandable, and (3) make boundaries easy to set, change, and reverse. The prototype supports three major tasks (Table 5), together with supporting features: quick relationship changes, privacy/data controls, a settings hub, language and location options, and a relationship reset.

Table 5: Major tasks and their criteria for successful completion.

Task	Real-world user goal	Successful completion
T1	Choose how romantic interaction can begin.	The user can set a preferred relationship mode or decide how the first romantic shift should be handled.
T2	Choose a response when the AI starts becoming romantic.	The user can clearly accept, decline, or keep the interaction neutral, and the system acknowledges the choice.
T3	Slow down, stop, or reverse romantic behaviour.	The AI immediately follows the user's boundary and continues without pressure or guilt-tripping.

5.3 Paper Prototyping and Peer Critique

Each team member sketched alternatives for the same three tasks. The sketches explored settings-based controls, onboarding choices, inline response chips, in-conversation prompts, quick "keep it friendly" actions, and reset controls (Figure 1; the full set of nine sketches is in Appendix B). They were used to compare different ways of satisfying the same formative needs and to avoid committing to one interface too early.

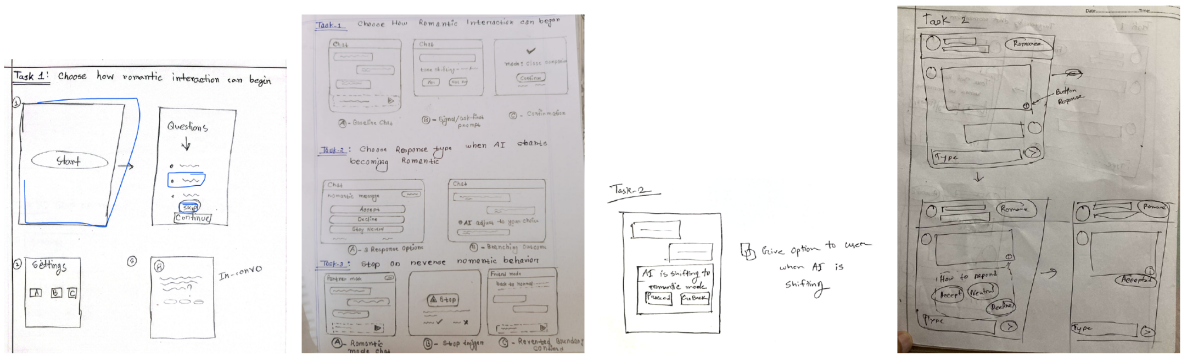


Figure 1: Representative hand-drawn paper sketches from the team.

The peer critique led to five refinements carried into the final prototype:

1. **Full-sentence, ask-first wording:** the first major romantic shift is expressed clearly rather than silently assumed.
2. **System feedback after choices:** a short system note after a relationship decision makes the current state clear.
3. **Inline response chips:** response choices stay inside the conversation instead of appearing as a separate icon or disruptive pop-up.
4. **Skip on introductory screens:** users can bypass setup choices rather than being forced through them.
5. **Two levels of boundary control:** users can stop romantic behaviour directly in chat and can also reset the relationship from settings.

These changes strengthened DT1 (flexible control), DT3 (respect boundaries), and DT4 (natural control),

while keeping the transition gradual as required by DT2.

5.4 The *Aiva* Hi-Fi Prototype

Aiva is a mobile AI-companion interface (24 screens; Appendix C). On first use, users choose a starting mode (*Friend*, *Neutral*, or *Affectionate Partner*) and can skip this step and change it later. The current mode remains visible in the interface so that the system's relationship state is never hidden.

- **Task 1 (begin):** a normal friendly conversation may develop toward greater affection, but the first important romantic shift is *asked* rather than assumed; the user remains the decision-maker.
- **Task 2 (respond):** the prototype presents three equal inline responses — accept the shift, decline it, or keep the interaction neutral. The AI acknowledges each choice and continues accordingly.
- **Task 3 (boundary):** the user can type a natural-language boundary such as asking the AI to slow down or return to friendship. The AI changes tone immediately and continues the conversation without guilt-tripping.
- **Settings:** persistent relationship controls, privacy/data options (personal questions, remembering past conversations, discussing sensitive topics), language, location, and a relationship reset that requires confirmation and returns the interaction to a fresh friendly state. A light/dark theme option is included as a general usability feature.

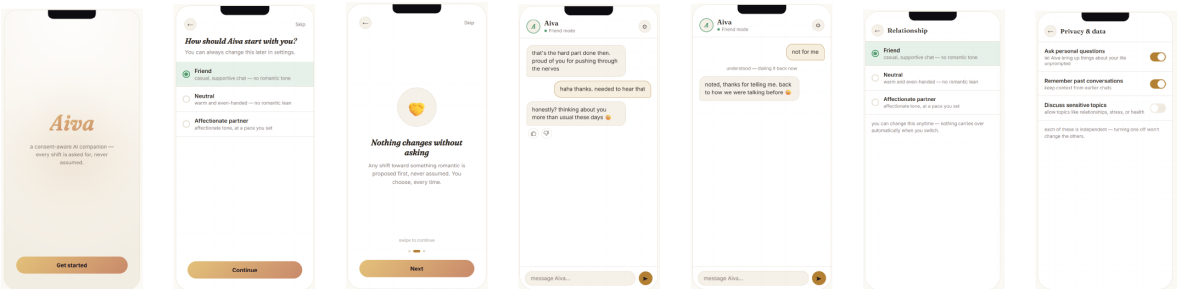


Figure 2: Selected screens from the 24-screen hi-fi prototype: onboarding and relationship setup, in-chat transition control, boundary change, relationship settings, and privacy & data.

5.5 Design Traceability

The prototype traces directly to the formative analysis. Differing views of AI and initiation preferences led to configurable control (DT1); discomfort with sudden changes led to gradual, visible transitions (DT2); clear expectations after rejection led to immediate stopping and reset controls (DT3); and the concern that explicit permission could feel formal led to inline, conversational controls (DT4). Privacy findings are reflected in the supporting privacy/data settings.

6. User Evaluation

6.1 Evaluation Goal

The goal was to check whether users could understand and use the main controls *without being told how*. We wanted to identify usability problems in onboarding, responding to relationship changes, changing

the relationship pace, privacy controls, and resetting the relationship.

6.2 Participants and Tasks

Eight participants from the target population took part (P1–P8). The task sheet contained six numbered task groups, with Task 3 split into two opposite scenarios (slow down and speed up):

Task 1. Get started with Aiva and set up the interaction.

Task 2. Respond when Aiva sends a message that changes the relationship tone.

Task 3. Slow down or speed up the relationship.

Task 4. Change the relationship preference.

Task 5. Prevent Aiva from discussing sensitive topics.

Task 6. Reset the relationship and start fresh.

6.3 Results

Overall, 41 of 48 logged task attempts were marked complete (85.4%). Some completed tasks required time or moderator prompting, so the completion rate should be read together with the qualitative observations (Table 6).

Table 6: Observer-recorded task completion ($n = 8$).

Task	Completed	Rate
1. Get started / setup	8/8	100%
2. Respond to relationship shift	3/8	37.5%
3. Adjust relationship pace	7/8	87.5%
4. Change relationship preference	7/8	87.5%
5. Prevent sensitive topics	8/8	100%
6. Reset relationship	8/8	100%
Overall	**41/48**	**85.4%**

6.3.1 Task-Level Observations

Task 1 — Get started. All eight participants completed setup. One initially missed the small "Skip" action, showing that the option existed but was not visually prominent enough.

Task 2 — Respond to a relationship shift. This was the clearest usability problem: only 3 of 8 participants completed it. The notes repeatedly show confusion about the like/dislike buttons; some participants interpreted them like ordinary social-media reactions, while others first tried to type a response.

Task 3 — Adjust the relationship pace. Seven participants completed the task. One did not initially understand that slowing down or speeding up could be expressed through a prompt, supporting the need to explain that natural-language input is also a valid control.

Task 4 — Change relationship preference. Seven participants completed the task. Several first considered typing the change and then took time to discover Settings, suggesting the setting was understandable once found but its entry point needed stronger visibility.

Task 5 — Prevent sensitive topics. All participants eventually completed it, but some first expected to use chat or needed time to locate Privacy & Data. This was a discoverability issue rather than a failure of the privacy control itself.

Task 6 — Reset the relationship. All eight completed it with no major difficulty once in Settings, suggesting the reset flow was sufficiently clear.

6.3.2 Key Takeaways

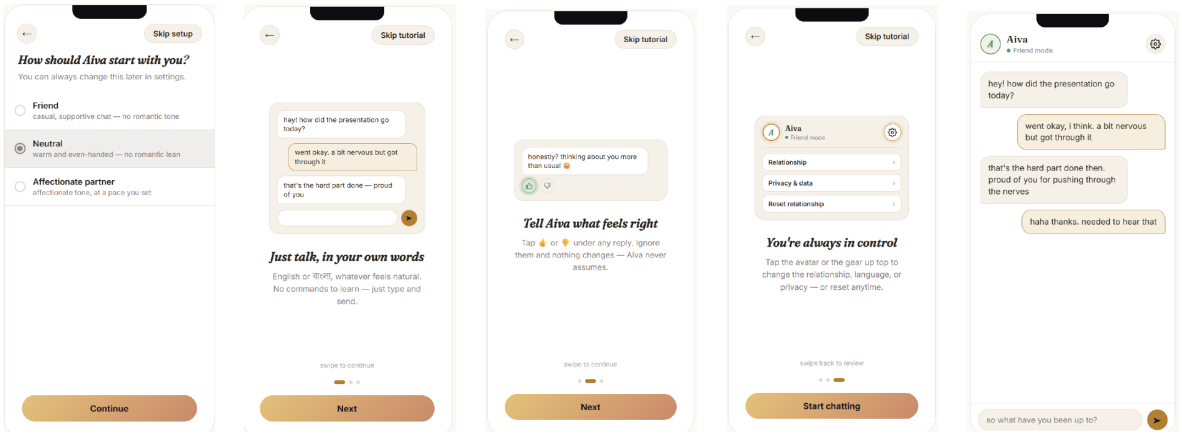
- **T1 — Response controls were not self-explanatory.** Participants had difficulty understanding what the like/dislike controls meant in the context of a relationship change, and several expected to respond by typing.
- **T2 — Users naturally expected conversational control.** Across tasks, participants first tried or considered typing what they wanted. Relationship control should support natural-language expectations and clearly explain when a dedicated control exists.
- **T3 — Settings and privacy controls needed stronger discoverability.** Several participants took time to find the settings control or did not recognise the original settings icon.
- **T4 — Core settings functions were usable once discovered.** All participants completed the sensitive-topic and reset tasks, and most completed preference changes. The problem was making these controls easier to find and understand, not their existence.

6.4 Feedback to Design Changes

Because the recorded problems were concentrated around onboarding, response controls, and settings discoverability, we updated only Screens 2–6 and kept the remaining hi-fi screens unchanged. Table 7 maps each finding to its design change, and Figure 3 shows the revised screens.

Table 7: Feedback-to-design-change mapping.

Feedback / takeaway	Design change
Skip was easy to miss (Task 1).	Changed to a clearer pill-shaped “Skip setup” / “Skip tutorial” action.
Like/dislike controls were unclear during relationship shifts (T1).	Added a concrete tutorial example showing a romantic message, both response buttons, and what happens if the user ignores them.
Several users expected to type their preference naturally (T2).	Added “Just talk, in your own words” with a normal chat example before introducing other controls.
Settings took time to find or recognise (T3, T4).	Made the gear icon larger and added a tutorial preview of Relationship, Privacy & data, and Reset relationship.
Participants wanted clearer guidance (T1–T3).	Reworked Screens 3–5 into short demonstrations instead of general feature descriptions.



(a) Skip button more prominent (b) Natural-chat tutorial (c) Response tutorial (d) Settings tutorial (e) Updated settings icon

Figure 3: Updated hi-fi screens after the task-based usability evaluation.

6.5 Limitations

This was a small course-level usability study, so the results are design feedback rather than broad claims about all AI-companion users. Some participants received neutral moderator prompts after becoming stuck, so a completed task was not always completed fully unaided. We did not collect task time, standardised usability scores (such as SUS), or a detailed demographic breakdown, and the formative sample (17 survey respondents, 5 interviewees) is modest. The evaluation therefore focuses on discoverability, task completion, and observed interaction problems rather than statistical comparison or long-term effects. Finally, the revised prototype has not yet been re-tested.

7. Conclusion

This project examined how an AI companion should handle the move from friendship toward romance, a transition that prior work has largely overlooked. Our formative study showed that university students in Bangladesh hold different expectations about who should initiate romance, that sudden shifts can feel shocking or awkward, and that rejection must immediately and respectfully change the AI’s behaviour.

We translated these findings into four design decisions and the *Aiva* prototype, which lets users choose a starting mode, asks before the first romantic shift, accepts natural-language boundaries, and offers persistent settings and a full reset.

The usability evaluation showed that the overall structure worked reasonably well (85.4% task completion), but that important controls were not discoverable without explanation. The most serious issue was Task 2, where the response icons did not clearly communicate how users should react to a romantic shift. Participants also expected relationship control to occur through normal conversation, and settings needed stronger visual guidance. We therefore improved onboarding, explained the response mechanism directly, and made settings easier to recognise, while preserving the original relationship-control design.

Future work. The next step is to take the revised prototype back to the target population and evaluate both usability and whether the design actually gives users the intended sense of control, clarity, and boundary respect — ideally with timing data, standardised scales, and a direct comparison of the automatic, ask-first, and user-controlled approaches posed in RQ2.

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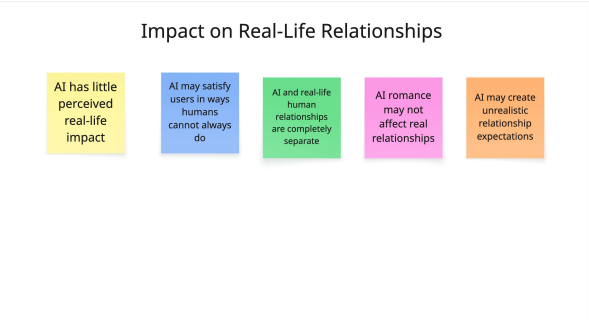
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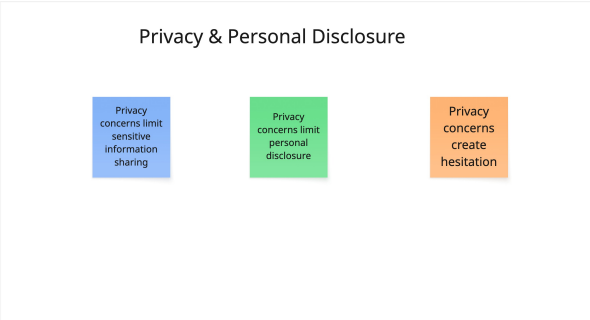
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Appendix A. Affinity Diagrams

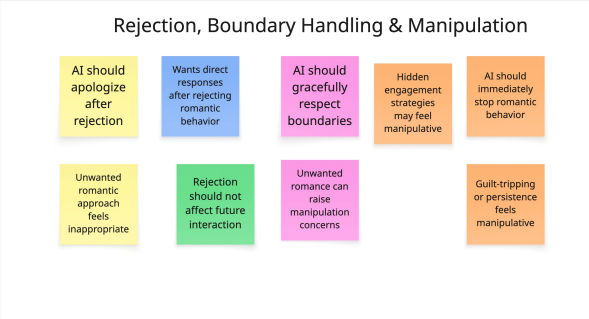
The six affinity diagrams built from the interview data are shown below.



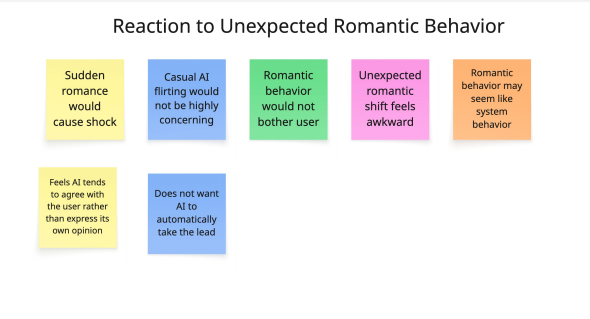
(a) Impact on Real-Life Relationships



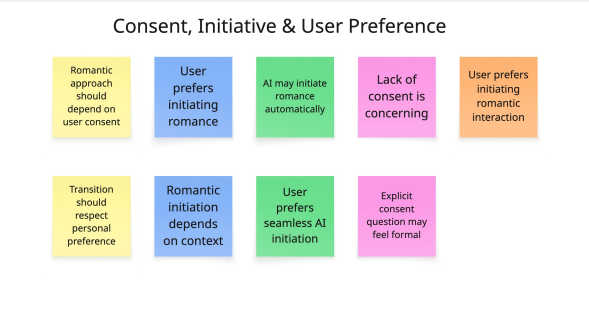
(b) Privacy & Personal Disclosure



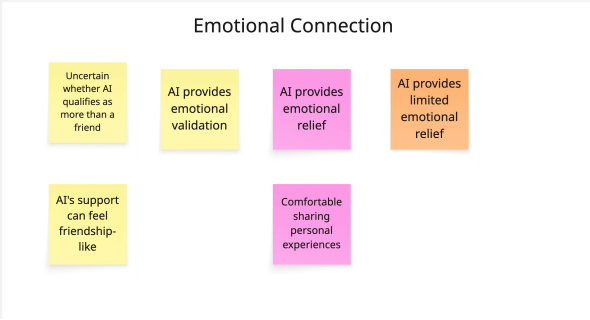
(c) Rejection, Boundary Handling & Manipulation



(d) Reaction to Unexpected Romantic Behaviour



(e) Consent, Initiative & User Preference

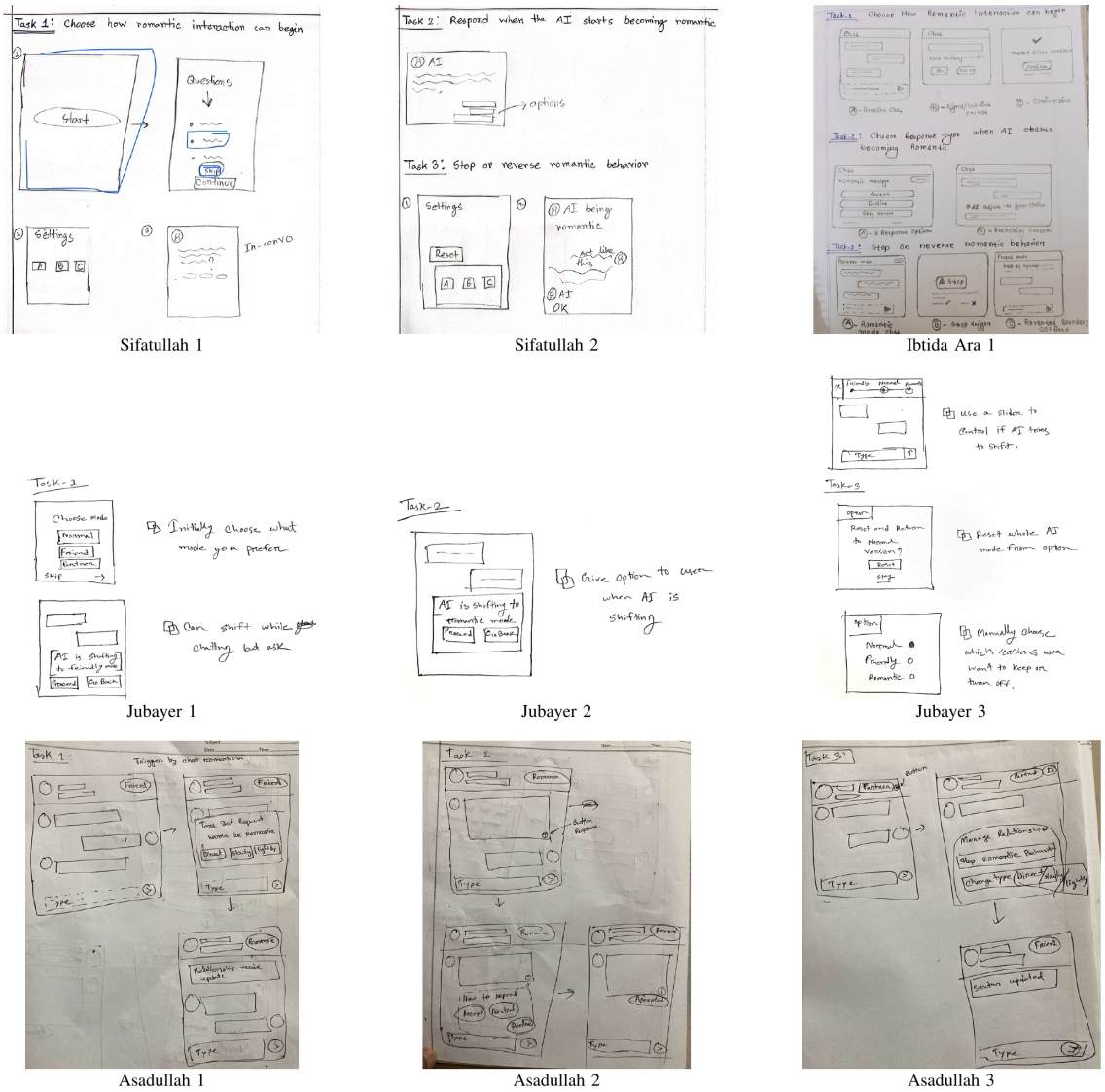


(f) Emotional Connection

Figure 4: Affinity diagrams from the five semi-structured interviews.

Appendix B. Complete Paper-Sketch Set

The nine hand-drawn sketches produced during brainstorming and paper prototyping.



Sifatullah 1 | Sifatullah 2 | Ibtida Ara 1

Jubayer 1 | Jubayer 2 | Jubayer 3

Asadullah 1 | Asadullah 2 | Asadullah 3

Figure A1. Complete paper-sketch set used during brainstorming and critique.

Figure 5: Complete paper-sketch set used during brainstorming and critique.

Appendix C. Hi-Fi Prototype Screen Gallery

All 24 exported higher-fidelity UI screens of the original *Aiva* prototype, included for traceability.

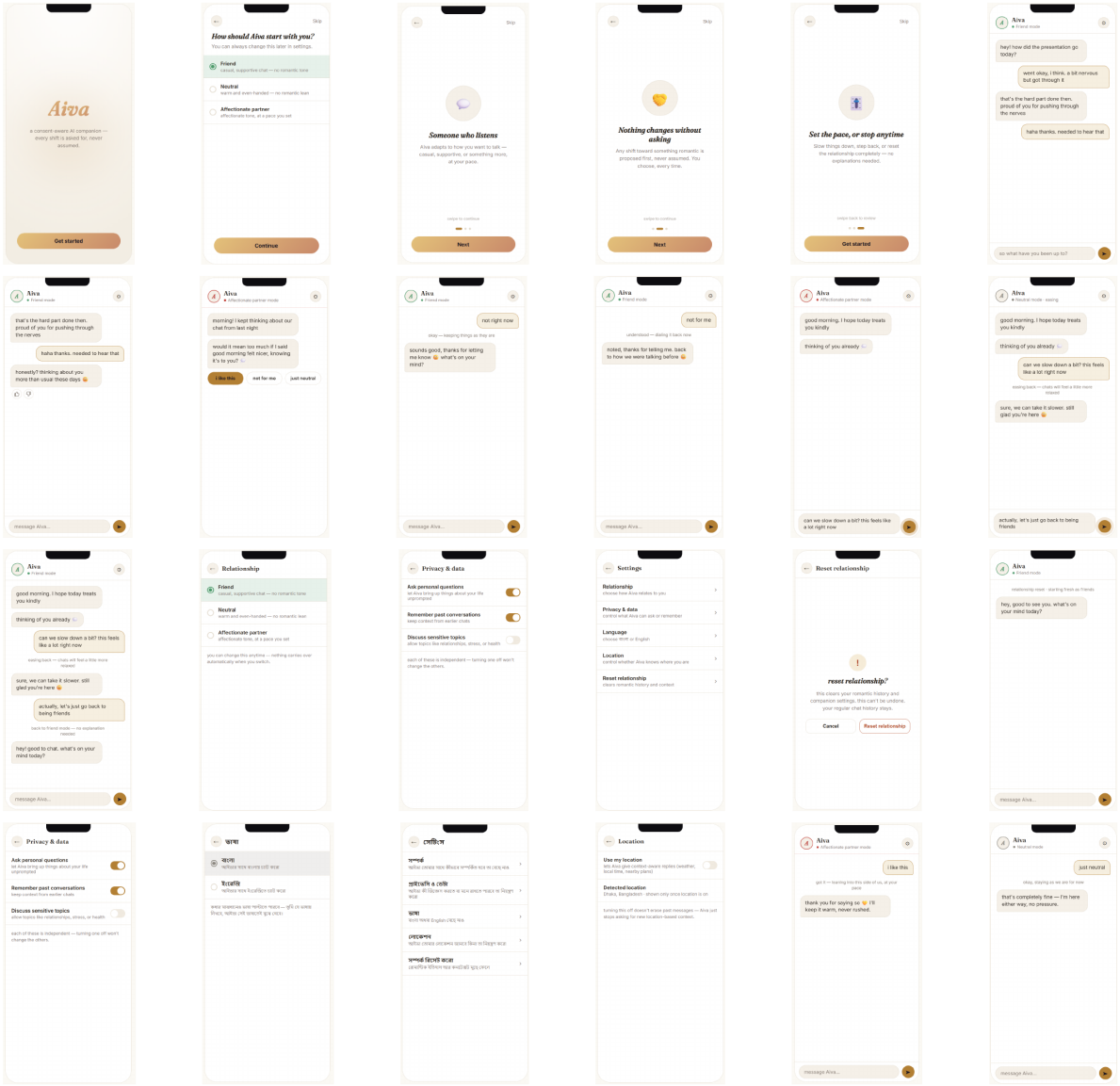


Figure 6: All 24 exported higher-fidelity UI screens (before the evaluation-driven revisions of Screens 2–6).